Lecture plan · for the lecturer

Recursion and induction

Programming · University · Lecture 2

Time
60 min

Materials
Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Explain the call stack.
- Argue termination.
- Prove by induction.

Activity: Be the stack

1. Warm-up (5 min): لماذا انهارت؟
2. Act (15 min): Five students act as frames for fact(4), passing results back up.
3. Trace (15 min): Pairs trace fib(4) and count calls.
4. Prove (20 min): Write the proof of fact(n) = n! together, then one for list length.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- What does each frame remember?
- Why is fib slow?
- What may you assume?

Exit ticket

- Write the inductive hypothesis for a recursive list sum.

Try it in your next lecture
Line up five students as stack frames for fact(4); each writes their n on a card and passes the result back, then the class writes the induction proof together.

Seminar sheet · for students

Recursion and induction

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Name _____ Class _____ Date _____

1 **In the inductive step, what do you assume?**

◯ The smaller call is correct ◯ The base case is wrong ◯ The function never returns

2 **Count the calls.**

Naive recursive fib(5), with fib(0) = 0 and fib(1) = 1, returns _____ and makes _____ calls in total.

Word bank: 5 · 8 · 9 · 15

3 **Which does a correct recursive function need? Choose all that apply.**

☐ A base case that is reachable ☐ A measure that decreases each call ☐ A global counter ☐ A correct combination of the smaller result

4 **In the inductive step you may assume the function is correct for the input you are proving.**

◯ True ◯ False

5 **Put the proof of fact(n) = n! in order.**

☐ Conclude (k+1) × k! = (k+1)!

☐ Expand fact(k+1) = (k+1) × fact(k)

☐ Assume fact(k) = k!

☐ Prove fact(0) = 1 = 0!

✎ **Which mental model of recursion do your weakest students hold, and how would you find out?**